Review (Matrix Product : Special Case)

Suppose $A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}$ is a $1 \times n$ matrix and $\mathbf{v} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$ is an $n \times 1$ vector.Then the product $A\mathbf{v}$ is defind as follows :

$$A\mathbf{v} = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 b_1 + a_2 b_2 + \cdots + a_n b_n \end{bmatrix}$$

The result is a 1×1 matrix.

Example.

$$\begin{bmatrix} 1 & -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} (1 \times 2) + ((-1) \times (-3)) + (0 \times 1) + (2 \times 1) \end{bmatrix} = \begin{bmatrix} 7 \end{bmatrix}$$

Example. The product of a 1×5 matrix and a 3×1 vector does not make sense.

$$\begin{bmatrix} * & * & * & * & * \end{bmatrix} \begin{bmatrix} * \\ * \\ * \end{bmatrix} = \text{☹}$$

• The product $A\mathbf{v}$ makes sense only if the number of columns of A is the same as the number of rows of $\mathbf{v}$.

Review...

The product of an $m \times n$ matrix and an $n \times 1$ vector

If $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$ Then :

$$A\mathbf{b} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = b_1\mathbf{a}_1 + b_2\mathbf{a}_2 + \cdots + b_n\mathbf{a}_n$$

Example.

$$\begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & -1 & -2 & 1 \\ 2 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -3 \\ -1 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} + (-3)\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} + (-1)\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} -6 \\ 6 \\ -3 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -5 \\ 4 \\ -1 \end{bmatrix}$$

Matrix Equation Form of a Linear System

Suppose we are given a linear system : $\begin{cases} -x_1 & +2x_2 & +x_3 & = 1 \\ 2x_1 & -4x_2 & & = -4 \\ -x_1 & +2x_2 & +3x_3 & = -1 \end{cases}$

1. We can write it in matrix form as follows :

$$\begin{bmatrix} -x_1 & +2x_2 & +x_3 \\ 2x_1 & -4x_2 & \\ -x_1 & +2x_2 & +3x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ -1 \end{bmatrix}$$

2. Then we "separate" the variables :

$$\begin{bmatrix} -x_1 \\ 2x_1 \\ -x_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ -4x_2 \\ 2x_2 \end{bmatrix} + \begin{bmatrix} x_3 \\ 0 \\ 3x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ -1 \end{bmatrix}$$

3. We can now rewrite it as:

$$x_1 \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -4 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ -1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} -1 & 2 & 1 \\ 2 & -4 & 0 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ -1 \end{bmatrix}$$

Matrix equation form of the linear system.

Review...

Suppose $\mathbf{b}, \mathbf{a}_1, ..., \mathbf{a}_p$ are vectors in $\mathbb{R}^n$. The following statements are equivalent (i.e., either they are all true or all false) :

1. The linear system whose augmented matrix is

$$\left[\begin{array}{cccc|c} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_p & \mathbf{b} \end{array}\right]$$

   is consistent.

2. The vector $\mathbf{b}$ is a linear combination of $\mathbf{a}_1, ..., \mathbf{a}_p$.

3. The vector equation

$$x_1\mathbf{a}_1 + \cdots + x_p\mathbf{a}_p = \mathbf{b}$$

   has a solution for $x_1, ..., x_p$.

4. If $A = \left[\begin{array}{cccc} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_p \end{array}\right]$ then the matrix equation $A\mathbf{x} = \mathbf{b}$ has a solution.

Review...

Therefore the following questions are essentially equivalent:

Question 1. Does the linear system $\begin{cases} -x_1 & +2x_2 & +x_3 & -x_4 & = 1 \\ 2x_1 & -4x_2 & & & = -4 \\ -x_1 & +2x_2 & +3x_3 & +2x_4 & = -1 \end{cases}$ have a solution?

Question 2. Is $\begin{bmatrix} 1 \\ -4 \\ -1 \end{bmatrix}$ a linear combination of $\begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -4 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$?

Question 3. Is $\begin{bmatrix} 1 \\ -4 \\ -1 \end{bmatrix}$ in Span$\left\{ \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -4 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} \right\}$?

Question 4. Does the matrix equation $\begin{bmatrix} -1 & 2 & 1 & -1 \\ 2 & -4 & 0 & 0 \\ -1 & 2 & 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ -1 \end{bmatrix}$ have a solution?

Review...

A linear system is consistent if and only if the rightmost column of its augmented matrix is not a pivot column.

From the above fact it follows that :

Guaranteed Existence of Solution

Suppose A is an $m \times n$ matrix, i.e.,

$$A = [\ \mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n\ ]$$

Where $\mathbf{a}_1, ..., \mathbf{a}_n$ are vectors in $\mathbb{R}^m$. Then the following statements are logically equivalent (i.e., they are either all true or all false) :

1. For every vector $\mathbf{b}$ in $\mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution.
2. Every vector $\mathbf{b}$ in $\mathbb{R}^m$ is a linear combination of $\mathbf{a}_1, ..., \mathbf{a}_n$.
3. The columns of A span $\mathbb{R}^m$.
4. A has a pivot position in every row.

Solution sets of linear systems

Today's goal :

- We have seen how to write the general solution of a linear system.
- Today we want to express the solution set of a linear system in two new ways:
  1. vector parametric form
  2. geometric form
- First we do this for a special case (the homogeneous systems) and then we extend our method to the general case (or the non-homogeneous systems).

Homogeneous system

A linear system is called homogeneous if the constants on the right hand side are all zeros.

Example. The linear system $\begin{cases} -x_1 & +x_2 & -3x_3 & -x_4 & = 0 \\ 2x_1 & -4x_2 & & +x_4 & = 0 \\ x_1 & +2x_2 & -3x_3 & -2x_4 & = 0 \end{cases}$ is homogenous.

The matrix equation of a homogeneous system is of the form $A\mathbf{x} = \mathbf{0}$.

The above system can be written as the matrix equation

$$\begin{bmatrix} -1 & 1 & -3 & -1 \\ 2 & -4 & 0 & 1 \\ 1 & 2 & -3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

As usual, we can solve any homogenous system using the R.E.F.

A homogeneous linear system is always consistent (because $x_1 = 0, x_2 = 0, \cdots$ is a solution).

Example

Solve the following linear system and write the solution in **vector parametric form**:

$$\begin{cases} -9x_1 & -2x_2 & +3x_3 & = 0 \\ 3x_1 & +x_2 & -x_3 & = 0 \\ 6x_1 & +x_2 & -2x_3 & = 0 \end{cases}$$

- First write the augmented matrix and apply the Row Reduction algorithm to obtain the R.E.F.

$$\left[\begin{array}{ccc|c} -9 & -2 & 3 & 0 \\ 3 & 1 & -1 & 0 \\ 6 & 1 & -2 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 1 & 0 & -\frac{1}{3} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

- Therefore x_1, x_2 are basic, x_3 is free, and the parametric form of the solution is :

$$\begin{cases} x_1 & & -\frac{1}{3}x_3 & = 0 \\ & x_2 & & = 0 \\ & & 0 & = 0 \end{cases} \quad \rightsquigarrow \quad \begin{cases} x_1 = \frac{1}{3}x_3 \\ x_2 = 0 \\ x_3 = \textit{free} \end{cases}$$

- In vector parametric form, we can write the solution as :

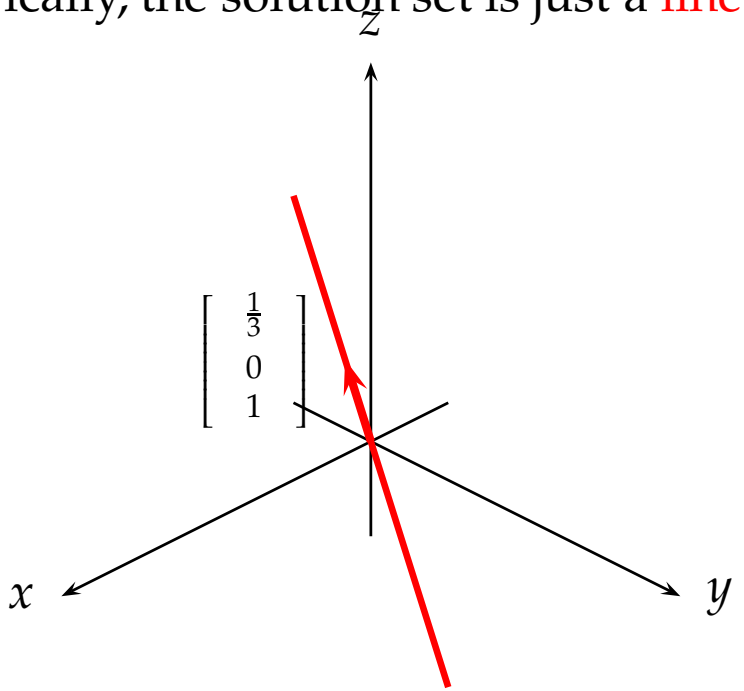
$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{3}x_3 \\ 0 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} \frac{1}{3} \\ 0 \\ 1 \end{bmatrix}$$

- Solution set: all vectors of the form $x_3 \begin{bmatrix} \frac{1}{3} \\ 0 \\ 1 \end{bmatrix}$.

  It follows that the solution set is spanned by <u>one</u> vector in $\mathbb{R}^3$, i.e., it is equal to

$$\text{Span}\left\{\begin{bmatrix} \frac{1}{3} \\ 0 \\ 1 \end{bmatrix}\right\}.$$

- Therefore, geometrically, the solution set is just a line that passes through the origin.



Example

Solve the following linear system

$$\begin{cases} 2x_1 - x_2 + 5x_3 = 0 \end{cases}$$

and write the solution in vector parametric form.

- The augmented matrix is $\left[\begin{array}{ccc|c} 2 & -1 & 5 & 0 \end{array}\right]$.

$$\left[\begin{array}{ccc|c} 2 & -1 & 5 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{5}{2} & 0 \end{array}\right]$$

  Therefore x_1 is basic, x_2, x_3 are free, and the general solution is given by

$$\begin{cases} x_1 - \frac{1}{2}x_2 + \frac{5}{2}x_3 = 0 \\ x_2 : free \\ x_3 : free \end{cases} \Rightarrow \begin{cases} x_1 = \frac{1}{2}x_2 - \frac{5}{2}x_3 \\ x_2 : free \\ x_3 : free \end{cases}$$

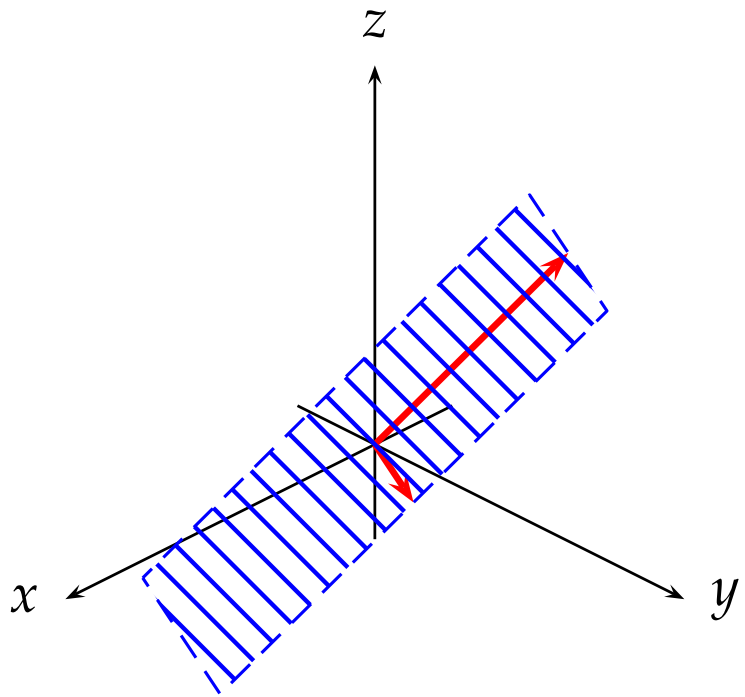
- The vector parametric form of the solution is obtained as follows :

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}x_2 - \frac{5}{2}x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -\frac{5}{2} \\ 0 \\ 1 \end{bmatrix}$$

- Therefore the solution is equal to $\text{Span}\left\{\begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{5}{2} \\ 0 \\ 1 \end{bmatrix}\right\}$

Geometric description of Span $\left\{ \begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{5}{2} \\ 0 \\ 1 \end{bmatrix} \right\}$:

- Geometrically, the vectors span a plane that passes through the origin.



- We saw that a homogeneous linear system is always consistent.
- Therefore a homogeneous linear system has :
  - a unique solution if every variable is basic.
  - infinitely many solutions if there is at least one free variable.
- The vector parametric form of the solution tells us that the solution set is equal to

$$\text{Span}\{\mathbf{v}_1, ..., \mathbf{v}_p\}$$

  for some vectors $\mathbf{v}_1, ..., \mathbf{v}_p$.
- If the homogeneous linear system has a unique solution then the solution set is $\text{Span}\{\mathbf{0}\} = \{\mathbf{0}\}$.

Nonhomogeneous systems

Example

Solve the linear system $\begin{cases} -x_1 + 2x_2 + x_3 = 1 \\ 2x_1 - 4x_2 = -4 \\ -x_1 + 2x_2 + 3x_3 = -1 \end{cases}$ and write the solution in **vector parametric form**.

The augmented matrix and its R.E.F. are :

$$\left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 1 & -2 & 0 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The parametric form of the general solution is :

$$\begin{cases} x_1 - 2x_2 = -2 \\ x_3 = -1 \\ 0 = 0 \end{cases} \rightsquigarrow \begin{cases} x_1 = 2x_2 - 2 \\ x_2 : \text{free} \\ x_3 = -1 \end{cases}$$

Again we write the solution in vector parametric form as follows:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_2 - 2 \\ x_2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2x_2 \\ x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \\ -1 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \\ -1 \end{bmatrix}$$

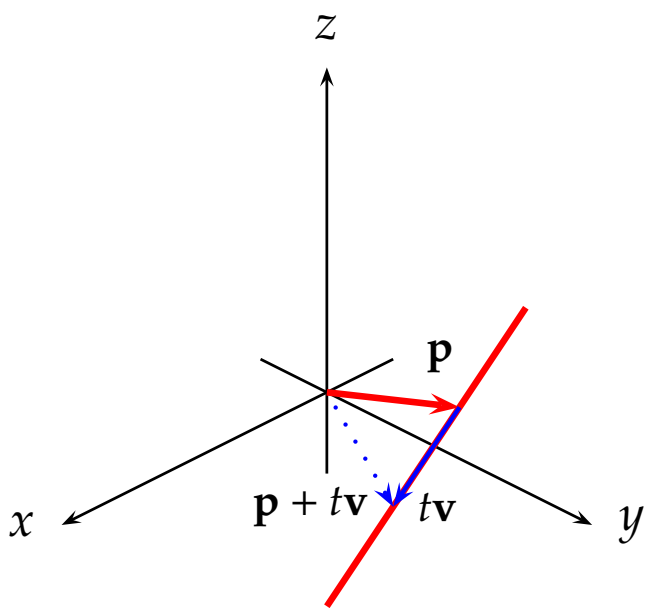
$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_2 - 2 \\ x_2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2x_2 \\ x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \\ -1 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

- If we set $\mathbf{p} = \begin{bmatrix} -2 \\ 0 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ then the solution set consists of all vectors $\mathbf{x}$ of the form

$$\mathbf{x} = \mathbf{p} + t\mathbf{v}$$

where $t = x_2$ can be any real number.

- Geometrically, we can describe the solution as a line that passes through $\mathbf{p}$ and is parallel to $\mathbf{v}$.



Example

Solve the linear system $\begin{cases} y_1 & -3y_2 & & -y_4 & = 2 \\ y_1 & +3y_2 & +y_3 & +y_4 & = 1 \\ 2y_1 & -6y_2 & +y_3 & & = 3 \end{cases}$

The augmented matrix and it R.E.F. are:

$$\left[\begin{array}{cccc|c} 1 & -3 & 0 & -1 & 2 \\ 1 & 3 & 1 & 1 & 1 \\ 2 & -6 & 1 & 0 & 3 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 2 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & -1 \end{array}\right] \text{ R.E.F.}$$

$$\begin{cases} y_1 = y_4 + 2 \\ y_2 = 0 \\ y_3 = -2y_4 - 1 \\ y_4 : \text{free} \end{cases}$$

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} y_4 + 2 \\ 0 \\ -2y_4 - 1 \\ y_4 \end{bmatrix} = \begin{bmatrix} y_4 \\ 0 \\ -2y_4 \\ y_4 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ -1 \\ 0 \end{bmatrix} = y_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

$A\mathbf{x} = \mathbf{b}$ vs. $A\mathbf{x} = \mathbf{0}$

Theorem

Suppose that the equation $A\mathbf{x} = \mathbf{b}$ is consistent (i.e., has a solution). Fix a solution $\mathbf{x} = \mathbf{p}$. The general solution of the equation $A\mathbf{x} = \mathbf{b}$ is of the form

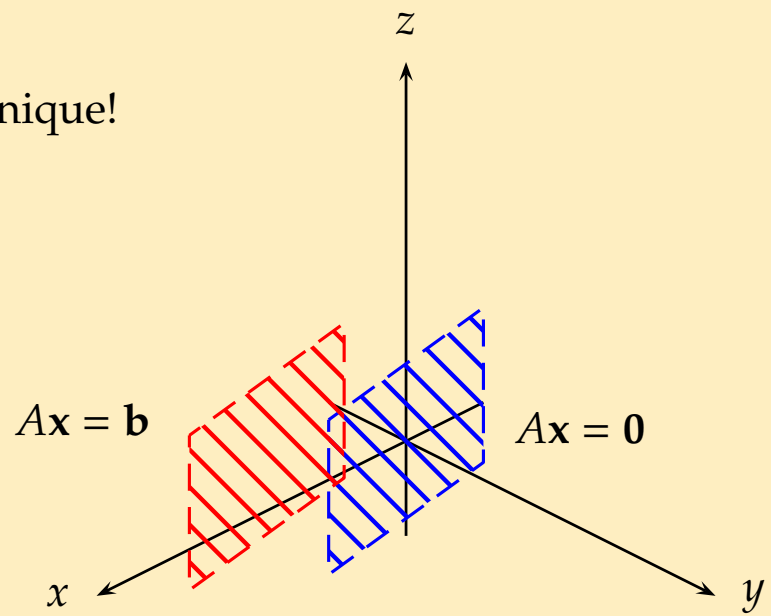
$$\mathbf{x} = \mathbf{p} + \mathbf{v}_h$$

where $\mathbf{v}_h$ is *any* solution of the homogeneous system $A\mathbf{x} = \mathbf{0}$.

$$A\mathbf{x} = \mathbf{b} \text{ and } A\mathbf{p} = \mathbf{b} \quad \Rightarrow \quad A(\mathbf{x} - \mathbf{p}) = A\mathbf{x} - A\mathbf{p} = \mathbf{b} - \mathbf{b} = \mathbf{0} \quad \Rightarrow \quad \mathbf{x} - \mathbf{p} = \mathbf{v}_h$$

Any such a vector $\mathbf{p}$ is sometimes called a particular solution of the nonhomogeneous linear system.

Remark. Note that $\mathbf{p}$ is not necessarily unique!



Example

The general solution of the linear system

$$\begin{cases} -x_1 & +2x_2 & -x_3 & = 0 \\ 2x_1 & -4x_2 & 0 & = 0 \\ -x_1 & +2x_2 & +3x_3 & = 0 \end{cases}$$

is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

We know that the values

$$x_1 = 3,\ x_2 = 2,\ x_3 = 2$$

form a solution for the linear system

$$\begin{cases} -x_1 & +2x_2 & -x_3 & = -1 \\ 2x_1 & -4x_2 & 0 & = 2 \\ -x_1 & +2x_2 & +3x_3 & = 7 \end{cases}$$

Write down the general solution of the latter linear system.

Solution.

$$\mathbf{x} = \mathbf{p} + \mathbf{v}_h \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$